

University of Chittagong
Department of Computer Science and Engineering
8th Semester B.Sc. (Engg.) Examination-2020
Course Code: CSE-813 Course Title: Distributed and Cloud Computing
Total marks: 52.5 Marks Time: 4.00 hours

[Answer any *three* questions from each of the *Section-A* and *Section-B*. A separate answer script must be used for Section-A and Section-B. Figures in the right-hand margin indicate full marks.]

Section-A

1. a) Define distributed and cloud systems? Give appropriate examples. 2+1
b) Differentiate tightly and loosely coupled systems with appropriate figures. 3.25
c) In what respect are distributed computing systems better than parallel processing systems? 2.5

2. a) How are location, relocation, and migration transparencies different from each other in distributed computing? Explain with examples. 3
b) Illustrate the commonly used techniques for implementing fault detection and recovery methods in a distributed operating system. 2.75
c) What are the design principles considered useful for better performance in a distributed system? 3

3. a) What is the primary motivation for the development of RPC? Draw a schematic diagram of implementation of the RPC mechanism and illustrate the operation in brief. 1.5+3
b) What is a stub? How are they generated? State their functionality and purpose. 1.25+3

4. a) What is causal consistency? Give an example of an application for which causal consistency is the most suitable consistency model. 1+2
b) Differentiate between weak consistency and release consistency. Which of the two will you prefer to use in the design of a DSM system? Give reasons for your answer. 2+1.5
c) Why does the simple LRU policy often used for replacing cache lines in a buffer cache not work well as a replacement policy for replacing blocks in a DSM system? 2.25

Section-B

5. a) What is the communication protocol used in RPC? Illustrate different steps of RRA protocol with an appropriate figure.
 - d) How can the RRA protocol be redefined for concurrent access to multiple servers? Illustrate with an appropriate figure. 2.75
 - c) Define callback RPC and lightweight RPC. 2
6. a) Draw a schematic diagram to represent a distributed shared memory (DSM). 1.75
 - b) What is false sharing? What should be done to minimize the false sharing problem? 1+2
 - c) Discuss the relative advantages and disadvantages of using the NRNMB, NRMB, RMB, and RNMB strategies in the design of a DSM system. 4
7. a) Draw an example explaining why it is difficult to achieve UNIX semantics in a distributed file system? 2.5
 - b) What are the general principles for designing distributed file systems? 4
 - c) How can the cache be validated? 2.25
8. a) Define cloud computing. List the pros and cons of cloud computing. 2
 - b) Assume your company wants to launch an e-commerce website. Which cloud services and deployment models will you consider for the website? Briefly explain. 4
 - c) What aspects of cloud computing make it useful for prognostic health management applications? 2.75

University of Chittagong
Department of Computer Science and Engineering
8th Semester B.Sc. (Engineering) Examination 2021
Course Title: Distributed and Cloud Computing
Course Code: CSE 813, Full Marks: 52.5, Duration: 4:00 Hours
Answer any three questions from each section.

[All parts of a question should be answered sequentially; figures in the right hand margin indicate full marks]

Section A

1. a) Define a distributed system. Mention a few real time examples of distributed systems. 1.75
b) Mention the advantages of a distributed computing environment over standalone applications. What are the disadvantages of distributed systems? 2.5
c) Define scalability. What are the challenges involved in designing scalable distributed systems? 2.5
d) Define Marshalling and Unmarshalling. What are the ACID properties? 2
2. Suppose that it is early 2005 and you are designing a web site allowing registered users to upload videos and anybody to download and watch them. You expect and hope that the site will be popular, so you use replication to handle the anticipated request rate. (Assume that, unlike a certain mainstream site whose name you can probably guess, no comments are associated with the videos.)
a) What are strong and weak consistencies? Describe the advantages and disadvantages of using each for this application, paying particular attention to any assumptions you must make about the users' behavior. 3
b) You are asked to modify your site so that it can be used to store and annotate video evidence for law enforcement. This requires using strong consistency; why? 1.75
c) Suppose that you use nine replicas to meet demand and quorums to ensure strong consistency. What are the constraints on the sizes of read and write quorums? Give an example of quorum processing involving a write, two reads, and then a write. Use the notation v_i to indicate version i of the data and suppose that at the start of your example, each replica contains version 3. Make sure that your diagram is easy to understand and that you explain any notation that you use. 4
3. a) Assume your company wants to launch an e-commerce website. Which cloud services and deployment models will you consider for the website? Briefly explain. 2.75
b) Draw an example explaining why it is difficult to achieve UNIX semantics in a distributed file system? 2

- c) What is false sharing? What should be done to minimize the false sharing problem? 2
- d) What is consistency? What are the different types of consistency that can be implemented with the DSM systems? 2
4. a) Define distributed system overhead (DSO). Explain how scalable parallel database system can solve computationally expensive real world problem? 4.25
- b) A class of database applications that look for hidden patterns in a group of data. What are those patterns and mention extraction methodologies. 3.50
- c) In cloud architecture, what are the different phases involved? 1

Section B

5. a) What is on-demand functionality? How is it provided in cloud computing? 1.75
- b) What is a cloud? Which platforms are used for large scale cloud computing? 1
- c) What are the different layers in cloud computing? Explain working of them. What are the differences between cloud computing and distributed computing? 3
- d) What is "EUCALYPTUS" in cloud computing? Why is it used? What are the different models for deployment in cloud computing? 3
6. a) Define concurrency control. Discuss in detail about deadlock and locking schemes in concurrency control 2.75
- b) Define time stamp ordering. What are the differences between validation phase and update phase 2
- c) What is distributed file system? What are the requirements of distributed file system (or) Give the design issues of distributed file system? 2
- d) What is Andrew File System? Define file service architecture of AFS? 2
7. a) What is round robin load balancing? 2
- b) Briefly describe about SQL and No-SQL databases and its comparative analysis 4.75
- c) What are the various types of MapReduce jobs? 2
8. a) What is Remote Procedure Call (RPC)? List out the design issues for RPC? 2.75
- b) Define distributed shared memory. Discuss on DSM with suitable illustrations. 3.5
- c) Consider an implementation of the Paxos algorithm where a leader waits for less than a majority of acceptors to answer OK to a Prepare or an Accept message before it proceeds to executing the next steps. Explain how such an implementation can break the Paxos guarantees. 2.5

University of Chittagong
Department of Computer Science and Engineering
Eight Semester B.Sc. Engineering Examination 2022
Course Code: CSE 513 Course Title: Distributed and Cloud Computing
Full Marks: 54 Time: 4 Hours

[Answer any **three** questions from each **Section A** and **B**; Separate answer script must be used for **Section -A** and **Section -B**. Figures in the right-hand margin indicate full marks.]

Section A

1. a) Define cloud computing. What is the difference between horizontal scaling and vertical scaling? Describe scenarios in which you will use each type of scaling. 5
b) Assume your company wants to launch an e-commerce website. Which cloud services and deployment models will you consider for the website? 2
c) What do you mean by multi-tenancy? 2
2. a) Cloud computing services like Amazon's EC2 assign users virtual machines (VMs) instead of allocating physical machines directly. Doing so provides at least three major benefits to Amazon. Explain what these three benefits are, giving a brief motivation for each one. 3
b) One common assumption in Paxos is that different leaders use different proposal numbers. A colleague shows you an implementation of Paxos where:
1. Different leaders can use the same proposal numbers, and
2. An acceptor rejects a Prepare or Accept message only if the proposal number in the message is strictly smaller than the largest proposal number that acceptor has seen (i.e., exactly as presented in class and in the course notes).
Is this Paxos implementation, correct? Find a counterexample or explain briefly why it is correct. 6
3. Suppose we wish to implement a transaction processing system that maintains ACID properties even in the presence of crashes. In event of a crash, any information stored on disk can be retrieved, but any data stored in memory will be lost.
Briefly describe one serious shortcoming of each of the following implementation :
a) The database is updated on disk with each transaction. 3
b) The database is kept in memory and on disk, with the copy on disk updated every 50 transactions. 3
c) The database is kept in memory. A log file is maintained on disk recording every transaction. 3
4. a) Identify two important limitations that MapReduce places upon the Map function so that the framework can hide failures from the programmer. 2
b) How does MapReduce handle "straggler" tasks that take longer than all of the others (e.g., perhaps they're running on a slower machine or one with buggy hardware)? 2
c) What is false sharing? What should be done to minimize the false sharing problem? 2
d) What is consistency? What are the different types of consistency that can be implemented with the DSM systems? Define. 3

Section B

5. a) Define distributed system overhead (DSO). Explain how scalable parallel database system can solve computationally expensive real world problem? 4.5
 b) A Class of database applications that look for hidden patterns in a group of data. What are those patterns? Mention extraction methodologies? 3.5
 c) Show an example of bitemporal relation. 1
6. a) Discuss in detail about deadlock and locking schemes in concurrency control. 3
 b) How do we increase the robustness of all-to-all heartbeating? 2
 c) What is Andrew File System? Define file service architecture of AFS? *→ show diagram* 2
 d) How do you design a group membership protocol? *show diagram* 2
7. a) Consider the following two transactions and the order in which all operations are processed. [r[X] means reading from an object X, w[X] means writing to an object X.] 4

Transaction T_0	Transaction T_1
$r[X]$ $w[X]$ $r[Y]$ $w[Y]$ commit	$r[X]$ $r[Y]$ commit

Do you believe this transaction will go off without a hitch? If not, what issue is raised by this transaction?

- b) Define concurrency control. Preventing isolation from being violated can be done in what ways? Explain. 3
 c) What is Deadlock detection in distributed systems? When do Deadlocks Occur? 2
8. a) What is Remote Procedure Call (RPC)? List out the design issues for RPC? 3
 b) Define distributed shared memory. How do you implement DSM over a message-passing network? 3
 c) Compare Invalidate and Update protocols. Which one is preferable and why? 3

↓
 store
 at a time
 jobs

↓
 at a time
 update

open one door
 clean functions
 network protocol
 transparent
 proxy sub.

Using section-wise separate answer-script, answer any three of the four given questions from each of the following two sections. Answers to parts of each question should appear contiguously in the answer-script. Write concisely, keeping your handwriting legible.

SECTION A

1. (a) [2½ marks] In general, high-performance computing (HPC), programmed using message passing, what is the technique used to provide fault tolerance?
(b) [2 marks] What advantage does the technique referred to in 1(a) give MapReduce over MPI for failure handling and recovery?
(c) [2½ marks] MapReduce and the Google File System (GFS) were designed to work well together. What important optimization in MapReduce is enabled by having GFS expose block replica locations via an API?
(d) [2 marks] Identify two important limitations that MapReduce places upon the Map functions so that the framework can hide failures from the programmer.
2. (a) [1½ marks] Define consensus algorithms. Differentiate between logical vs. physical concurrency.
(b) [1½ marks] What is linearizability? Point out the method to ensure serializability.
(c) [3 marks] What is atomic commit protocol? Explain the two-phase commit protocol for nested transactions.
(d) [3 marks] What is the primary goal of failure detection in a distributed system? Briefly explain why achieving perfect and instant failure detection is practically impossible in an asynchronous network.
3. (a) [3 marks] Consider a simple server that carries out client requests without accessing other servers. Explain why it is generally not possible to set a limit on the time taken by a server to respond to a client request. What should the server do, to execute requests within a bounded time?
(b) [3 marks] How do you ensure the security in DFS? Briefly remark on the various message ordering paradigms used in distributed systems.
(c) [3 marks] What is the purpose of wait-for-graph (wfg)? Give an example of wfg. What are the conditions that a deadlock detection algorithm must satisfy?
(d) [1½ marks] Compare message passing and distributed shared memory approaches.
(b) [1½ marks] What is false sharing? What should be done to minimize the false sharing problem?
(c) [3 marks] Briefly explain the different types of ordering of multicasts messages in overlapping groups.
(d) [3 marks] What are venus and vice processes? How does AVS deal with the risk that callback messages may be lost?

SECTION B

5. (a) [2½ marks] List two distinct benefits and two significant drawbacks of adopting cloud computing for enterprises. Keep your answer within four crisp and strictly relevant sentences.
(b) [2½ marks] Explain, in three to four crisp and strictly relevant sentences, the fundamental concept of virtualization in cloud computing. Point out two key benefits that virtualization provides to cloud service providers in terms of resource management.
(c) [4 marks] Consider a remote procedure call (RPC)-based system where a client communicates with a server located across the network. Illustrate how marshalling and unmarshalling work during the invocation process. Evaluate how communication paradigms (message-passing, RPC/RMI etc.) affect system performance and reliability for time-sensitive applications.
6. (a) [3 marks] Explain how concurrency in distributed systems differs fundamentally from concurrency in centralized systems. Analyze how issues like partial failure and lack of global clock impact the design of concurrent operations in distributed environments. Keep your answer within a few crisp and strictly relevant sentences.

- (b) [3 marks] A mid-sized e-commerce company, currently operating on-premise, experiences highly fluctuating traffic with significant spikes during seasonal sales and promotional events. They are considering migrating their existing monolithic application to the cloud. Analyze the trade-offs between Platform as a Service and Infrastructure as a Service models for this company. Your analysis should focus on:
 - Initial development speed and flexibility
 - Operational overhead and management complexity
 - Vendor lock-in risk
 - Scalability during peak traffic
 Based on your analysis, which model would you initially recommend for their primary application and why?
- (c) [3 marks] The above-mentioned company determines that 80% of its workload is predictable, requiring 5 virtual machines (VMs) running 24/7. The remaining 20% is highly variable, spiking to require an additional 10 VMs for 4 hours/day, 7 days a week, during peak periods. Assume an IaaS provider charges:
 - On-demand instance: \$0.10/VM-hour
 - Reserved instance (purchased for full month, 24/7): \$0.05/VM-hour
 i. Calculate the monthly cost for running their predictable workload using reserved instances.
 ii. Calculate the monthly cost for running their variable workload using on-demand instances.
 iii. Calculate the total estimated monthly IaaS cost for this setup.
7. (a) [3 marks] Explain why Paxos cannot tolerate f failures with less than $2f + 1$ nodes.
(b) [3 marks] Consider an implementation of the Paxos algorithm where a leader waits for less than a majority of acceptors to answer OK to a Prepare or an Accept message before it proceeds to executing the next steps. Explain how such an implementation can break the Paxos guarantee.
(c) [3 marks] For (any two of) the following scenarios, indicate which replication scheme you would use: either primary-backup or Paxos.
 - i. A lock server for a distributed filesystem that supports buying and selling stocks on the Dhaka stock exchange.
 - ii. Storing data on data nodes in a distributed filesystem that stores your collection of documentary movies.
 - iii. A login server (accepts username/password, tells whether OK, and allows users to change passwords) used to support both of the above services, and many others.
8. (a) [3 marks] In the context of distributed systems, what is ACID? Tell us what each letter stands for and give a single crisp, concise sentence explaining what each letter means.
(b) [3 marks] Hasan, Sumon, and Aman decide to try out rowing crew.

Unfortunately, there's only `while (!at destination) { recover_strength(); grab_one_bottle(); grab_one_bottle(); grab_one_bottle(); grab_one_bottle(); row_the_boat(); row_the_boat(); row_the_boat(); drop_one_bottle(); drop_one_bottle(); drop_one_bottle(); }`

As a way to arbitrate access to the `bottles`, Hasan proposes that the oars should be placed in the center of the boat, and every rower will follow a simple protocol:

Is it possible that this plan will lead to deadlock? Keep your answer within six crisp and strictly relevant sentences.
- (c) [3 marks] For each of the following two statements, take a position (or evaluate) as to whether it is true or not. Then present a sentence or two to explain your position/evaluation.
 - i. The Chandy-Lamport global snapshot algorithm works correctly for non-FIFO channels.
 - ii. In a system with n processes, the Chandy-Lamport snapshot algorithm will always show that at least $n-1$ channels are empty.

- (c) [3 marks] Briefly explain the different types of ordering of multicasts messages in overlapping groups.
- (d) [3 marks] What are venus and vice processes? How does AFS deal with the risk that callback messages may be lost?

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- (b) [3 marks] Hasan, Suman, and Aman decide to try out rowing crew.

Unfortunately, there's only one boat left that can seat three people (it leaks a bit), and there are only four *boitthas*. A rower must have two *boitthas* in order to row, or else the boat will go in circles.

```
while (!at destination){
    recover_strength();
    grab_one_boittha();
    grab_one_boittha();
    row_the_boat();
    row_the_boat();
    drop_one_boittha();
    drop_one_boittha();
}
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As a way to arbitrate access to the *boitthas*, Hasan proposes that the oars should be placed in the center of the boat, and every rower will follow a simple protocol:

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING,
UNIVERSITY OF CHITTAGONG
8TH SEMESTER BSc ENGINEERING EXAMINATION 2024
CSE 813: DISTRIBUTED AND CLOUD COMPUTING
MARKS: 54 TIME: 4 HOURS

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- (b) [3 marks] A mid-sized e-commerce company, currently operating on-premise, experiences highly fluctuating traffic with significant spikes during seasonal sales and promotional events. They are considering migrating their existing monolithic application to the cloud. Analyze the trade-offs between Platform as a Service and Infrastructure as a Service models for this company. Your analysis should focus on:

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Explain why in a sentence or two for each.

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